



GRACE COLLEGE OF ENGINEERING
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MULLAKKADU, THOOTHUKUDI - 628 005

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Unit- III Game Playing and CSP

Prepared By,

Mrs.R.Janani, AP/AI&DS

Unit- III Game Playing and CSP

PART A

1. How can minmax also be extended for game of chance?

In a game of chance we can add extra level of chance nodes in g tree. These nodes have successors which are the outcomes of random element. The minmax algorithm uses probability P attached with chance node d_i based value. Successor function $S(N, d_i)$ give moves from position N for outcome d

2. What are the 2 parts of Landscape?

i. Location defined by the state. ii. Elevation defined by the value of the heuristic cost function (or) objective function.

3. Define Global minimum.

If elevation corresponds to cost, then the aim is to find the lowest valley is called global minimum.

4. Define Global Maximum.

If elevation corresponds to an objective function, then the aim is to find the highest peak is called global maximum.

5. Define Hill Climbing search.

It is a loop that continually moves in an increasing value direction (i.e.) up hill and terminates when it reaches a “peak” where no neighbor has a higher value.

6. List some drawbacks of hill climbing process.

Local maxima: A local maxima as opposed to a goal maximum is a peak that is lower than the highest peak in the state space. Once a local maxima is reached the algorithm will halt even though the solution may be far from satisfactory.

7. Define Plateaux.

A plateau is an area of the state space where the evaluation fn is essentially flat. The search will conduct a random walk.

8. What are the variants of hill climbing?

i. Stochastic hill climbing ii. First choice hill climbing iii. Simulated annealing search iv. Local beam search v. Stochastic beam search

9. Define annealing.

Annealing is the process used to harden metals (or) glass by heating them to a high temperature and then gradually cooling them, thus allowing the material to coalesce into a low energy crystalline state.

10. Define simulated annealing.

This algorithm, instead of picking the best move, it picks a random move. If the move improves the situation, it is always accepted.

11. What is the advantage of memory bounded search techniques?

We can reduce space requirements of A^* with memory bounded algorithm such as IDA^* & SMA^* .

12. What is Genetic Algorithms?

Genetic Algorithm is a variant of stochastic beam search in which successor states are generated by combining two parent states, rather than by modifying a single state.

13. Define Online Search agent. Agent operates by interleaving computation and action (i.e.) first it takes an action, and then it observes the environment and computes the next action.

14. What are the things that agent knows in online search problems?

a. Actions(s) b. Step cost function $C(s, a, s')$ c. Goal TEST(s)

15. Define CSP.

Constraint Satisfaction problem (CSP) is defined by a set of variables $X_1, X_2, \dots, X_n$ and set of constraints $C_1, C_2, \dots, C_m$.

16. Define Successor function.

A value can be assigned to any unassigned variable, provided that does not conflict with previously assigned variables.

17. What are the types of constraints?

There are 5 types, a. Unary constraints relates one variable. b. A binary constraint relates two variables. c. Higher order constraints relate more than two variables. d. Absolute constraints. e. Preference constraints.

18. Define constraint propagation.

It is the general term for propagating (i.e.) spreading the implications of constraints on the variable on to other variable.

19. Define Cycle cut set.

The process of choosing a subset S from variables [CSP] such that the constraint graph becomes a tree after removal of S . S is called a cycle cut set.

20. Define Tree decomposition.

The constraint graph is divided into a set of connected sub problems. Each sub problem is solved independently and the resulting solutions are then combined. This process is called tree decomposition.

21. Define Alpha beta pruning.

Alpha beta pruning eliminates away branches that cannot possibly influence the final decision.

22. Give a precise formulation of the following CSP in terms of variables, domain and constraints: there are 5 professors and 10 classrooms, a list of classes to be offered and a list of possible time slots for classes Each professor has a set of classes that he or she can teach.

To formulate the given problem as a Constraint Satisfaction Problem (CSP), we define the following:

1. Variables:

- **Professors:** Each professor is a variable. Let P_1, P_2, P_3, P_4, P_5 represent the 5 professors.
- **Classes:** Each class to be offered is a variable. Let $C_1, C_2, \dots, C_n$ represent the classes (where n is the number of classes to be offered).
- **Classrooms:** Each class needs a classroom, so we define $R_1, R_2, \dots, R_{10}$ as the 10 classrooms.

23. What do you mean by constraint propagation?

Constraint propagation refers to the process of reducing the domains of variables in a constraint satisfaction problem (CSP) by using the constraints to eliminate impossible or inconsistent values.

It involves systematically updating the domains of variables based on the constraints, to narrow down the possible solutions and make the problem easier to solve. For example, if a variable has a value that violates a constraint, that value is removed from its domain. This process helps reduce the search space and can lead to earlier detection of inconsistency or solution discovery.

24. What are the 3 levels in describing knowledge-based agent?

The three levels in describing a knowledge-based agent are:

1. **Knowledge Level:** This level focuses on what the agent knows about the world and its environment. It describes the agent's knowledge and the reasoning it can perform to solve problems, make decisions, and achieve goals.
2. **Symbol Level:** This level refers to how knowledge is represented within the agent. It deals with the symbols and structures used to represent the knowledge, such as logical statements, rules, or frames.

3. **Implementation Level:** This level describes how the knowledge and reasoning processes are implemented in the agent's architecture. It deals with the computational methods and algorithms that allow the agent to use and manipulate the knowledge.

These levels provide a hierarchical framework for understanding how knowledge is acquired, represented, and used in a knowledge-based agent.

25. Justify why we cannot use traditional min max for games with an element of chance, such as backgammon.

Traditional **minimax** cannot be directly used for games with an element of chance, such as **backgammon**, because **minimax** assumes a deterministic environment where the outcome of every action is predictable. In games with chance, such as backgammon, the outcomes of moves are influenced by random events (e.g., dice rolls).

In such games, the agent's actions do not guarantee specific outcomes, and the future state of the game is not fully determined by the player's decisions alone. This introduces uncertainty, which requires the use of a different approach, such as **expectimax**, which accounts for the probabilistic nature of the game by incorporating the expected value of each possible outcome, based on the chance events.

Thus, **minimax** is unsuitable because it does not handle the randomness inherent in games like backgammon.

26. Class scheduling: There is a fixed number of professors and classrooms, a list of classes to be offered, and a list of possible time slots for classes. Each professor has a set of classes that he or she can teach. Give a precise formulation.

To formulate the **class scheduling problem** as a **Constraint Satisfaction Problem (CSP)**, we define the following:

1. Variables:

- **Professors:** Let $P_1, P_2, \dots, P_m$ represent the m professors.
- **Classes:** Let $C_1, C_2, \dots, C_n$ represent the n classes to be offered.
- **Classrooms:** Let $R_1, R_2, \dots, R_k$ represent the k classrooms.
- **Time Slots:** Let $T_1, T_2, \dots, T_l$ represent the l time slots.

2. Domains:

- **Professors:** The domain of each professor's variable is the set of classes they can teach. For each professor P_i , the domain is the set of classes they are eligible to teach (e.g., $D(P_i)$ is the list of classes for that professor).
- **Classrooms:** The domain for each class variable is the set of available classrooms (i.e., $\{R_1, R_2, \dots, R_k\}$).
- **Time Slots:** The domain for each class variable is the set of available time slots (i.e., $\{T_1, T_2, \dots, T_l\}$).

3. Constraints:

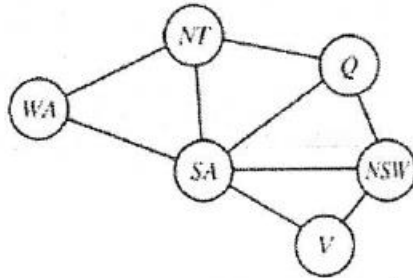
- **Professor-Class Assignment:** A professor can only teach the classes they are qualified for, so each professor can only be assigned classes from their eligible set.
- **Classroom Availability:** Each classroom can only be assigned to one class at a time. This means no two classes can be scheduled in the same classroom at the same time.
- **Time Slot Assignment:** Each class must be scheduled at a specific time slot, ensuring no two classes are scheduled at the same time in the same classroom.
- **Professor Schedule:** A professor cannot teach more than one class in the same time slot, so each professor's schedule must have distinct time slots for their assigned classes.

This formulation captures the key aspects of the problem using variables, domains, and constraints.

PART-B

1. Explain the perfect decisions in game playing with example.
2. Explain in detail about the minmax algorithms and how it works for tic-tac-toe.
3. Explain about alpha beta pruning with examples.
4. Explain the Optimal decisions in game.
5. Explain the components of Monte-Carlo tree search with example.
6. Discuss stochastic games with example.
7. What is CSP with examples.
8. Briefly explain the Constraint propagation with example.
9. What is backtracking search for and explain with an example.
10. Explain the constraint satisfaction problem and the variations on constraint satisfaction problem with example.
11. Explain short notes on Monte-Carlo search.
12. Define local consistency. What are the different types of local consistency? Explain any two.
13. What is the significance of knowledge representation ? Mention the desirable properties of knowledge representation.
14. How does alpha beta search algorithm differ from MinMax algorithm? Analyse.

Consider the map coloring problem with 6 variables and three colors (red, green and blue). The constraint graph for the problem is given below. How does backtracking search solve the given problem? What are the heuristics used to improve the efficiency of the search? How are failures detected early in backtracking? Can Breadth first search be applied to the above problem? State reasons for your answer.



15. What are partially observable games? How are they solved in a deterministic environment?
16. Formulate the Map-Coloring problem as CSP and explain with an example.
17. The missionaries and cannibals' problem is usually stated as follows.
Three missionaries and three cannibals are on one side of a river, along with a boat that can hold one or two people. Find a way to get everyone to the other side without ever leaving a group of missionaries in one place outnumbered by the cannibals in that place.
 - i) Formulate the problem precisely, making only those distinctions necessary to ensure a valid solution. Draw a diagram of the complete state space.
 - ii) Implement and solve the problem optimally using an appropriate search algorithm. Is it good idea to check for repeated states?
 - iii) why do you think people have a hard time solving this puzzle, given that the state space is so simple?